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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

Extra 1 - Semantics of Propositional Logic

1.1. Let p and q be the propositions "I play the lottery" and "I win the *jackpot*", respectively. Express each of the following compound propositions as a sentence in natural language (English or Portuguese):

(a) $\neg p$
(b) $p \wedge q$
(c) $\neg p \vee \neg q$
(d) $p \rightarrow q$
(e) $q \rightarrow p$
(f) $\neg p \rightarrow \neg q$

1.2. Using truth tables or the semantic equivalences given in class (boolean algebra), show the following equivalences between propositions.

(a) $\neg(p \rightarrow q) \Leftrightarrow p \wedge \neg q$
(b) $\neg p \rightarrow q \Leftrightarrow \neg q \rightarrow p$
(c) $p \rightarrow (q \rightarrow r) \Leftrightarrow q \rightarrow (p \rightarrow r)$
(d) $(p \rightarrow q) \rightarrow p \Leftrightarrow p$
(e) $(p \vee q) \rightarrow r \Leftrightarrow (p \rightarrow r) \wedge (q \rightarrow r)$

1.3. Assigning truth values to p, q, and r investigate how many of the disjunctions $p \vee \neg q$, $\neg p \vee q$, $q \vee r$, $q \vee \neg r$, $\neg q \vee \neg r$ can be simultaneously true.

1.4. For the following formulas indicate, **justifying**, if they are tautologies, satisfiable or contradictions. You can start by simplifying the formulas using semantic equivalences. For each of the formulas that you think is not a tautology or a contradiction, indicate an assignment that makes the formula true and another that makes the formula false.

(a) $(p \rightarrow (q \vee r)) \vee (r \rightarrow \neg p)$;
(b) $p \rightarrow (q \rightarrow \neg(\neg q \rightarrow \neg p))$;
(c) $(p \rightarrow q) \rightarrow (\neg p \rightarrow \neg q)$;
(d) $p \rightarrow (q \rightarrow (p \vee q))$;
(e) $(q \rightarrow r) \rightarrow ((\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r))$;
(f) $(p \vee q) \rightarrow (q \rightarrow (p \wedge q))$;
(g) $(p \rightarrow q) \wedge \neg((q \rightarrow s) \rightarrow (p \rightarrow s))$;
(h) $(p \rightarrow q) \wedge (\neg r \rightarrow (q \vee (\neg p \wedge r)))$;

(i) $(p \vee q) \to (p \wedge q)$
(j) $(p \vee \neg q) \wedge (\neg p \wedge q)$
(k) $((p \vee q) \wedge r) \leftrightarrow ((p \wedge r) \vee (q \wedge r))$

1.5. Justify the truth or falsity of each of the following statements where p, q, r and s are propositional variables.

(a) $\models (p \wedge \neg p) \to (p \wedge \neg r)$.
(b) $\models (q \vee \neg r) \to (q \wedge \neg q)$.
(c) $\{(p \to q) \vee r, ((p \to q) \vee r) \to \neg r, (p \to q) \to (q \to r)\} \models \neg q$.
(d) $\{(p \vee q) \to r)\} \models (p \to r) \vee (q \to r)$.

1.6. Justify the truth or falsity of each of the following statements, where Γ and Σ represent sets of formulas and $\phi, \psi, \theta, \gamma$ represent formulas of propositional logic. If the statement is false, you must give a counterexample indicating particular values of the sets or formulas for which the statement is false. If the statement is true you must justify using the definitions.

(a) $\models (\theta \vee \phi) \to (\neg\theta \to \phi)$.
(b) $\{\phi \to \psi, \psi \to \phi\} \models (\phi \vee \psi) \to (\phi \wedge \psi)$.
(c) $\{\psi \to (\neg\phi \vee \gamma), \neg\gamma\} \models \neg\phi \to \neg\psi$.
(d) If $\phi, \psi \models \theta \to \gamma$ and $\phi \models \theta$ then $\phi, \psi \models \gamma$.

1.7. For each of the following formulas determine an equivalent formula in disjunctive normal form and another in conjunctive normal form.

(a) $\neg(p \to (\neg(q \wedge (\neg p \to q))))$;
(b) $p \to (\neg q \to \neg(p \to q))$;
(c) $\neg(p \to (\neg(q \wedge (\neg p \to q))))$;
(d) $(p \to q) \to (\neg p \to \neg q)$;
(e) $((p \to q) \to p) \to p$;
(f) $p \to \neg(q \to \neg(p \to q))$;
(g) $(p \to q) \wedge (\neg r \to (q \vee (\neg p \wedge r)))$;
(h) $(p \to q) \wedge (\neg q \to (p \vee \neg q))$;
(i) $p \to (q \to \neg(\neg q \to \neg p))$;
(j) $p \to (q \to (p \wedge q))$;
(k) $(p \to q) \to (\neg p \to \neg q)$;
(l) $((p \vee q) \to (r \wedge s)) \to (p \vee \neg r)$;
(m) $(p \to q) \wedge \neg((q \to s) \to (p \to s))$;

1.8. Formalize the following arguments, by choosing appropriate propositions and writing sets of formulas to represents the entire text. Then assume that every formula before "Thus" is an hypothesis and every formula after "Thus" is a conclusion and show whether: hypothesis $\models$ conclusion, or hypothesis $\not\models$ conclusion.

(a) "If Carlo won the competition, then either Mario came second or Sergio came third. Sergio didn't come third. Thus, if Mario didn't come second, then Carlo didn't win the competition."
(b) "If you play and you study you'll pass the exams, while if you play and don't study you won't pass. Thus, if you play, either you study and you'll pass the exams, or you don't study and you won't pass."